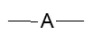
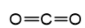
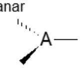
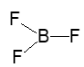
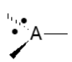
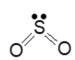
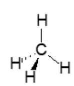
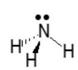
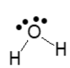
Drawing structure of molecules & polyatomic ions

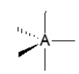
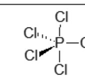
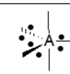
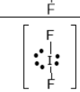
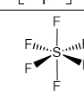
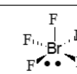
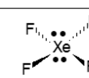
Method

1. Identify central atom
2. Arrange side atoms symmetrically
3. Distribute the charges
   - (+) on electro-positive
   - (—) on electronegative
4. Draw bonds to ensure side atoms have octet / duplet
5. Draw lone pair of electrons on central atom and check no. of electrons around central atom
6. Dative bond
   - if central atom cannot expand octet, convert double bond to dative bond
   - if side atoms have octet/ duplet w/o bonds, form dative bonds
   - if side atoms do not have electrons, form dative bonds
7. Erase charges on atoms; add bracket and charge outside bracket

NH_3	CO_3^{2-}	O_3	H_3O^+
PCl_5	SO_4^{2-}	N_3^-	NH_4^+
ClO_2	NO_2^-	NO_3^-	IF_4^-

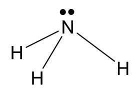
VSEPR

No of regions of e⁻ density	No. of bond pairs	No. of lone pairs	Molecular Geometry / Shape	Examples	Bond angle
2	2	0	linear 	CO_2 	180°
3	3	0	trigonal planar 	BF_3 	120°
	2	1	bent 	SO_2 	< 120°
4	4	0	tetrahedral 	CH_4 	109.5°
	3	1	trigonal pyramidal 	NH_3 	~107°
	2	2	bent 	H_2O 	~105°

No of regions of e⁻ density	No. of bond pairs	No of lone pairs	Molecular Geometry / Shape	Examples	Bond angle
5	5	0	trigonal bipyramidal 	PCl_5 	120° or 90°
	4	1	see–saw 	SF_4 	N.A. (not in syllabus)
	3	2	T–shape 	ClF_3 	90°
	2	3	linear 	IF_2^- 	180°
6	6	0	octahedral 	SF_6 	90°
	5	1	square pyramidal 	BrF_5 	< 90°
	4	2	square planar 	XeF_4 	90°

Application of VSEPR

Example 1

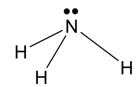


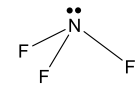
State and explain the shape and bond angle about the N atom in NH_3

- The N atom has _____ bond pairs and _____ lone pairs
- NH_3 has a ________________________________ shape
- The bond angle is ____________
- This is to ___________________________ between electron pairs about the N atom, where repulsion of

  ____

Example 2



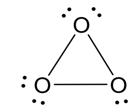


Explain why the bond angle in NH_3 is greater than the bond angle in NF_3

- F is _______ electronegative than H
- F attracts the bond pair of electrons closer to itself in NF_3 compared to H in NH_3.
- Therefore, the bond pair of electrons are closer together about N in _________
- There is greater ____________ between the ______________ of electrons about N in NH_3, hence the bond angle in NH_3 is greater.

Example 3

molecule **X**



Explain why molecule **X** would be highly unstable

- The ideal O-O-O bond angle is ___________,

  whereas the O-O-O bond in molecule **X** is __________(______)
- This causes the bond pair of electrons to be __________________
- Therefore, there is additional _________________ between the

  ____________________________ about O in molecule **X**, making molecule **X** highly unstable.

Structure and Bonding

Giant ionic lattice

- ionic bonding refers to the electrostatic attraction between __________________ and _________________

ionic bond strength

- __________________________________ is a measure of ionic bond strength

$$|\text{Lattice energy}| \propto \left|\frac{q_+ \cdot q_-}{r_+ + r_-}\right|$$

- factors affecting ionic bond strength are…

  ①

  ②

Giant metallic lattice

- metallic bonding refers to the electrostatic attraction between a lattice of __________ ____________ and the sea of _________________________

metallic bond strength

- factors affecting metallic bond strength are…

  ①

  ②

Giant molecular structure

- substances with giant molecular structures are made up of atoms held together by ________ ______________________
- essential examples are: _________ , ___________ , _______________

covalent bond strength

- factors affecting covalent bond strength are…

  ①

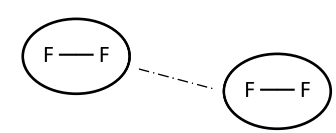
  ②

  ③

Simple molecular structure

- substances with simple molecular structures are made up of discrete molecules which have ___________________________ ________________________________
- examples are Ar, HC*l*, Br_2, A*l*C*l*$_3$
- during melting and boiling of these substances, it is the weak intermolecular forces of attraction that are overcome, **not** the strong covalent bonds.

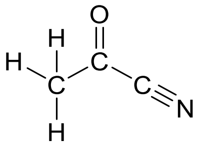
Label the covalent bond and the intermolecular forces of attraction (IMF)



Further classification of bonds

π and σ bonds

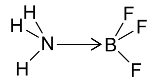
➢ Count all π and σ bonds



π bond: ___ (side-on overlap) σ bond: ___ (head-on overlap)

dative covalent bond

➢ Explain why NH_3 can form a dative bond with BF_3



- B in BF_3 contains ________________________ to accept the ______________________ from N in NH_3.

ionic bond with covalent character

➢ Draw to show the extent of polarisation/ distortion of electron cloud of the anion

Note:
The electron cloud of Cl^- ion is distorted to such a great extent that the $Al—Cl$ bond is covalent

➢ factors affecting extent of the polarisation of the electron cloud are…

① ______________ of cation (∝ charge density)

② ______________ of anion (∝ size of anion)

polar bonds
a.k.a. covalent bond with ionic character

➢ Draw to show any dipole moment

δ− δ+
F — C*l*

F — Br

F — F

F — I

Intermolecular forces of attraction (IMF)

id-id
instantaneous dipole – induced dipole

➢ factors affecting the strength of id-id are…

① no. of electron, electron cloud size

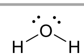
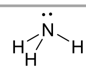
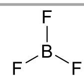
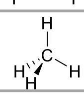
② ________________ for interaction (branch vs straight chain)

➢ By considering structure and bonding, explain why CH_4 has a lower boiling point than C_2H_6

- CH_4 and C_2H_6 are both covalent substances with ________________________ structure. There exists ____ (id-id) interactions between CH_4 molecules and between C_2H_6 molecules.
- C_2H_6 has ________________________________ compared to CH_4. Hence more energy is required to overcome the stronger the id-id in C_2H_6 than in CH_4.

pd-pd
permanent dipole – permanent dipole

Pd-pd interactions exists between polar molecules. Deduce which the following molecules are polar

structure	polar	non-polar
O=C=O		
		
		
		
		

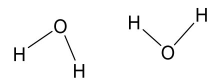
H-bond

➢ conditions for the formation of hydrogen bonds are…

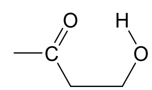
① $^{\delta+}$H atom bonded to ______

② lone pair (l.p.) of electrons on _______

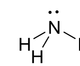
➢ Illustrate the **intermolecular** H-bonds formed between two H_2O molecules



➢ Illustrate the **intramolecular** H-bonds formed within the molecule shown below



➢ Deduce the average number of H-bonds (extensiveness) formed per molecule in NH_3



no. of H atom bonded to N/O/F = ____

no. of l.p. of electrons on N/O/F = ____

avg. no of H-bond per molecule: ____

Note:

- *intramolecular H-bond (if possible) is favored over intermolecular H-bond*
- *strength of H-bond is affected by size of dipole moment on H and N/O/F*

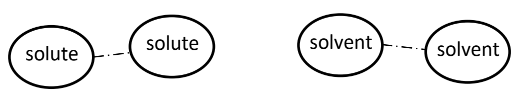
Melting/ Boiling

➢ State the interactions to be broken/ overcome during the melting/ boiling of the following substances

CH_4	
CH_2O	
C (diamond)	
H_2O	
NaC*l*	

Solubility

➢ When is it favorable for a solute to be dissolved in a solvent?



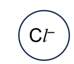
It is favorable when the energy released from the ____________________________________ interactions is more than the energy required to overcome the ____________________________________ interactions

Note:
Must state the specific interactions e.g. pd-pd / ionic bonds

➢ State the type of interaction between solute and solvent

solute	solvent	interaction
CH_4	CCl_4	
CH_4	CH_3Cl	
CH_2O	CH_3Cl	
CH_2O	H_2O	
NaC*l*	H_2O	

➢ Draw the interaction between Cl^- with H_2O



Note:

- $|\Delta H_{hydration}|$ *is a measure of the strength of ion-dipole interactions*
- $|\Delta H_{hydration}|$ ∝ charge density

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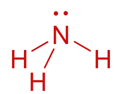
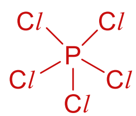
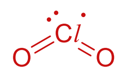
NH_3



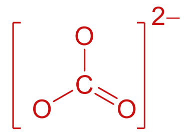
PCl_5



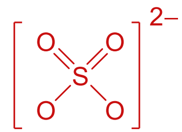
ClO_2



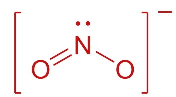
CO_3^{2-}



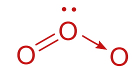
SO_4^{2-}



NO_2^-

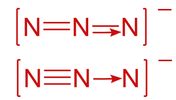


O_3

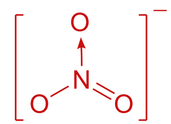


N_3^-

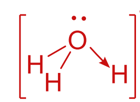
or



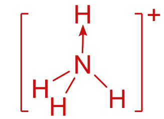
NO_3^-



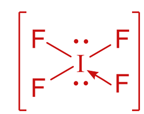
H_3O^+



NH_4^+



IF_4^-



VSEPR

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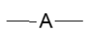
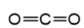
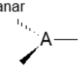
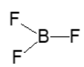
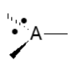
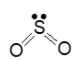
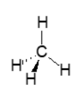
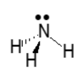
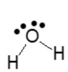
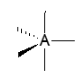
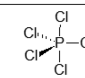
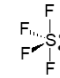
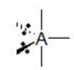
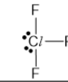
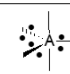
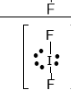
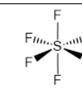
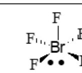
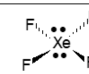
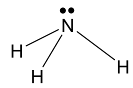
2	2	0	linear 	CO_2 	180°
3	3	0	trigonal planar 	BF_3 	120°
	2	1	bent 	SO_2 	< 120°
4	4	0	tetrahedral 	CH_4 	109.5°
	3	1	trigonal pyramidal 	NH_3 	~107°
	2	2	bent 	H_2O 	~105°

No of regions of e⁻ density	No. of bond pairs	No of lone pairs	Molecular Geometry / Shape	Examples	Bond angle
5	5	0	trigonal bipyramidal 	PCl_5 	120° or 90°
	4	1	see–saw 	SF_4 	N.A. (not in syllabus)
	3	2	T–shape 	ClF_3 	90°
	2	3	linear 	IF_2^- 	180°
6	6	0	octahedral 	SF_6 	90°
	5	1	square pyramidal 	BrF_5 	< 90°
	4	2	square planar 	XeF_4 	90°

Application of VSEPR

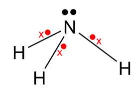
Example 1

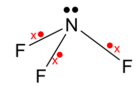


State and explain the shape and bond angle about the N atom in NH_3

- The N atom has __3__ bond pairs and __1__ lone pairs
- NH_3 has a __trigonal pyrimidal__ shape
- The bond angle is __107°__
- This is to __minimise repulsion__ between electron pairs about the N atom, where repulsion of __lone pair – bond pair > bond pair – bond pair__

Example 2



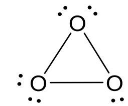


Explain why the bond angle in NH_3 is greater than the bond angle in NF_3

- F is __more__ electronegative than H
- F attracts the bond pair of electrons closer to itself in NF_3 compared to H in NH_3.
- Therefore, the bond pair of electrons are closer together about N in __NH_3__
- There is greater __repulsion__ between the __bond pair__ of electrons about N in NH_3, hence the bond angle in NH_3 is greater.

Example 3

molecule **X**



Explain why molecule **X** would be highly unstable

- The ideal O-O-O bond angle is __105°__, whereas the O-O-O bond in molecule **X** is __smaller__ (__60°__)
- This causes the bond pair of electrons to be __closer together__
- Therefore, there is additional __repulsion__ between the __bond pair of electrons__ about O in molecule **X**, making molecule **X** highly unstable.

Structure and Bonding

Giant ionic lattice

- ionic bonding refers to the electrostatic attraction between cations and anions

ionic bond strength

- magnitude of lattice energy is a measure of ionic bond strength

$$|\text{Lattice energy}| \propto \left|\frac{q_+ \cdot q_-}{r_+ + r_-}\right|$$

- factors affecting ionic bond strength are…
  1. charge of cation and anion
  2. ionic radii of cation and anion

Giant metallic lattice

- metallic bonding refers to the electrostatic attraction between a lattice of positively ions and the sea of delocalised electrons

metallic bond strength

- factors affecting metallic bond strength are…
  1. no. of valence electrons
  2. charge density of cation

     charge density $\propto \frac{q_+}{r_+}$

Giant molecular structure

- substances with giant molecular structures are made up of atoms held together by strong covalent bonds
- essential examples are: graphite, diamond, silicon dioxide (SiO_2)

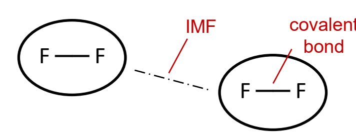
covalent bond strength

- factors affecting covalent bond strength are…
  1. no. of bonds/ no. of bonding electrons
  2. effectiveness of orbital overlap
  3. polarity of bond

Simple molecular structure

- substances with simple molecular structures are made up of discrete molecules which have weak intermolecular forces of attraction (IMF).
- examples are Ar, HC*l*, Br_2, $AlCl_3$
- during melting and boiling of these substances, it is the weak intermolecular forces of attraction that are overcome, **not** the strong covalent bonds.

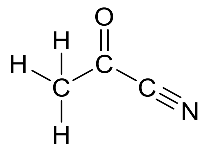
Label the covalent bond and the intermolecular forces of attraction (IMF)



Further classification of bonds

π and σ bonds

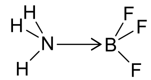
- Count all π and σ bonds



π bond: 3 (side-on overlap) σ bond: 7 (head-on overlap)

dative covalent bond

- Explain why NH_3 can form a dative bond with BF_3



- B in BF_3 contains vacant low-lying orbitals to accept the lone pair of electrons from N in NH_3.

ionic bond with covalent character

- Draw to show the extent of polarisation/ distortion of electron cloud of the anion

Note:
The electron cloud of Cl^- ion is distorted to such a great extent that the Al — Cl bond is covalent

- factors affecting extent of the polarisation of the electron cloud are…

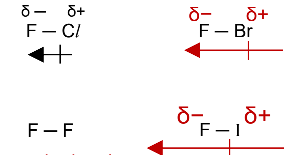
① polarising power of cation (∝ charge density)

② polarisability of anion (∝ size of anion)

polar bonds

a.k.a. covalent bond with ionic character

- Draw to show any dipole moment



non-polar bond *most polar bond*

Intermolecular forces of attraction (IMF)

id-id

instantaneous dipole – induced dipole

- factors affecting the strength of id-id are…

① no. of electron, electron cloud size

② surface area for interaction (branch vs straight chain)

- By considering structure and bonding, explain why CH_4 has a lower boiling point than C_2H_6

• CH_4 and C_2H_6 are both covalent substances with simple molecular structure. There exists weak instantaneous dipole – induced dipole (id-id) interactions between CH_4 molecules and between C_2H_6 molecules.

• C_2H_6 has a larger and more polarisable electron cloud compared to CH_4. Hence more energy is required to overcome the stronger the id-id in C_2H_6 than in CH_4.

pd-pd

permanent dipole – permanent dipole

Pd-pd interactions exists between polar molecules. Deduce which the following molecules are polar

structure	polar	non-polar
O=C=O		✓ dipoles cancel out
H_2O	✓	
NH_3	✓	
BF_3		✓ dipoles cancel out
CH_4		✓

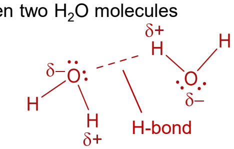
H-bond

- conditions for the formation of hydrogen bonds are…

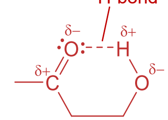
① $^{δ+}$H atom bonded to N/O/F

② lone pair (l.p.) of electrons on N/O/F

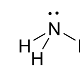
- Illustrate the **intermolecular** H-bonds formed between two H_2O molecules



- Illustrate the **intramolecular** H-bonds formed within the molecule shown below



- Deduce the average number of H-bonds (extensiveness) formed per molecule in NH_3



no. of H atom bonded to N/O/F = 3

no. of l.p. of electrons on N/O/F = 1

avg. no of H-bond per molecule: 1

Note:

- *intramolecular H-bond (if possible) is favored over intermolecular H-bond*
- *strength of H-bond is affected by size of dipole moment on H and N/O/F*

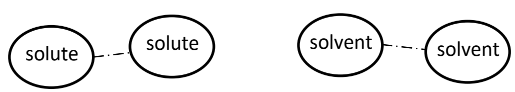
Melting/ Boiling

- State the interactions to be broken/ overcome during the melting/ boiling of the following substances

CH_4	id-id
CH_2O	pd-pd (and id-id)
C (diamond)	strong covalent bonds
H_2O	H-bond (and id-id)
NaCl	strong ionic bonds

Solubility

- When is it favorable for a solute to be dissolved in a solvent?



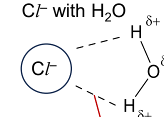
It is favorable when the energy released from the solute-solvent interactions is more than the energy required to overcome the solute-solute + solvent-solvent interactions

Note:
Must state the specific interactions e.g. pd-pd / ionic bonds

- State the type of interaction between solute and solvent

solute	solvent	interaction
CH_4	CCl_4	id-id
CH_4	CH_3Cl	id-id
CH_2O	CH_3Cl	pd-pd
CH_2O	H_2O	H-bond
NaCl	H_2O	ion-dipole

- Draw the interaction between Cl^- with H_2O



ion-dipole interaction

Note:

- $|\Delta H_{hydration}|$ *is a measure of the strength of ion-dipole interactions*
- $|\Delta H_{hydration}|$ ∝ charge density